

Modem Harika

3rd Year Undergraduate
Department of Mechanical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology, Kanpur	6.3/10
2023	TSBIE(XII)	Narayana Junior College, Kothapet	97.9%
2021	TS SSC(X)	Sri Chaitanya School,L.B.Nagar	10/10

Scholastic Achievements

- Secured **All India Rank 16316** in **JEE Advanced 2023** among the 1.89 Lakh shortlisted candidates.
- Secured **All India Rank 12671** in **JEE Mains 2023** among 1.36 million candidates.
- Secured **State Rank 823** in Telangana state engineering entrance test **TS EAMCET 2023** among 1.95 Lakh candidates.

Key Projects

- ShiftGuard10: Robust Image Classification** (Mar'26)
Course Project - EE708, IIT Kanpur
 - Developed a modified ResNet50 architecture optimized for low-resolution (32×32) images by refining initial layers to preserve spatial detail.
 - Implemented Manifold Mixup and stochastic transformations to improve model robustness and prevent overfitting on small-scale data.
 - Utilized OneCycleLR scheduling and AdamW optimization to achieve "Super-Convergence," ensuring faster and more stable training.
 - Achieved a 0.744 Macro F1-score, demonstrating balanced classification accuracy across 10 diverse categories.
- Applications of Large Language Models** (Dec'24 - Jan'25)
Association of Mechanical Engineers, IIT Kanpur
 - Developed a system that translates text from images into English using OCR and LLMs, bridging language barriers.
 - Designed a tool to summarize key themes from user search queries, making research more efficient.
 - Implemented a question-answering module to provide precise, context aware responses to user queries.
 - Developed and tuned encoder-decoder models using LSTMs/RNNs for sequence prediction tasks.
 - Summarized large PDF documents using LangChain tools like MapReduce and RefineChain.
- Vibration Analysis & PID Control of a Washing Machine** (Feb'26 - Mar'26)
Course Project - ME354, IIT Kanpur
 - Developed an SDOF mathematical model and derived transfer functions using Laplace-domain analysis to study vibration response and system stability.
 - Designed and tuned a PID controller achieving **94% reduction** in vibration amplitude, significantly improving stability and settling time.
 - Built and validated a Python-based simulation framework to compare controlled vs. uncontrolled system performance against theoretical predictions.

Technical skills

- Programming Languages:** C, C++, Python
- Software:** Autocad, MS Excel
- Libraries & ML Frameworks :** Pandas, NumPy, Tensorflow, Keras
- Developer Tools:** Git, LaTeX, VS Code

Positions of Responsibility

- Senior Executive (Public Relations)** , Udghosh, IIT Kanpur (Oct'24)
 - Led guest outreach and successfully invited high-profile personalities to the fest.
 - Coordinated with guests for scheduling, invitations, and smooth conduct of sessions.
 - Helped in writing and sending formal invites and managing guest communications. Worked with a team to ensure all guest-related activities ran smoothly during the fest.
- Organizer (Show Management)**, Udghosh, IIT Kanpur (Sept'25 - Oct'25)
 - Coordinated effectively with a team of heads, senior executives to ensure the seamless execution of the Festival.
 - Designed creative layouts and activity modules, enhancing participant engagement and overall visual presentation.
 - Supervised setup and decoration using handmade craft models, improving event aesthetics and organization quality.
 - Developed and implemented strategies for the decoration of event related areas utilising the innovative ideas of various members.

Relevant Courses

Fundamentals of Computing Machine Learning, Coursera (online) Thermodynamics Fluid Mechanics Engineering Graphics Nature and Properties of Materials	Linear algebra and Differential Equations Robotics Mechanics of Solids Dynamics Theory of Machines and Mechanism Primary and Secondary Manufacturing Process
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